Home Assignment – Linux Shell & Python Fundamentals

Week 1

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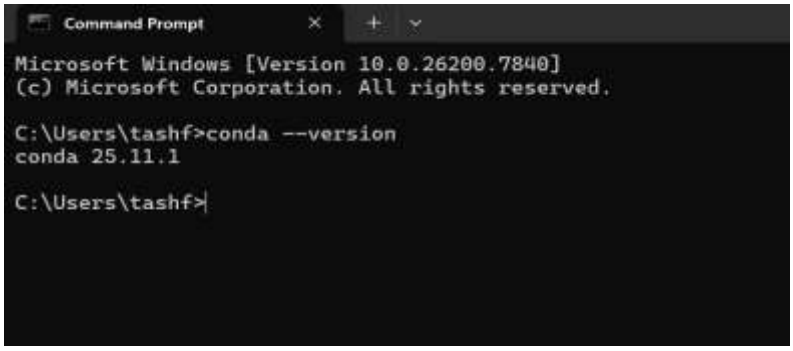
NIAI AI/ML Training

March 01, 2026

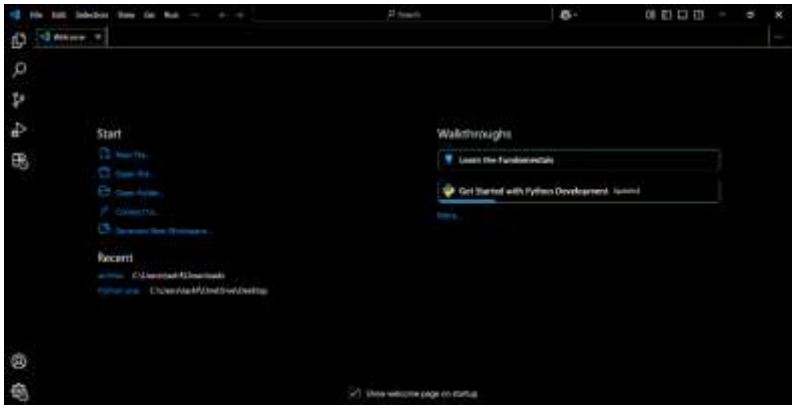
Part 1 – Software Installation

Task 1 – Installation

- Anaconda

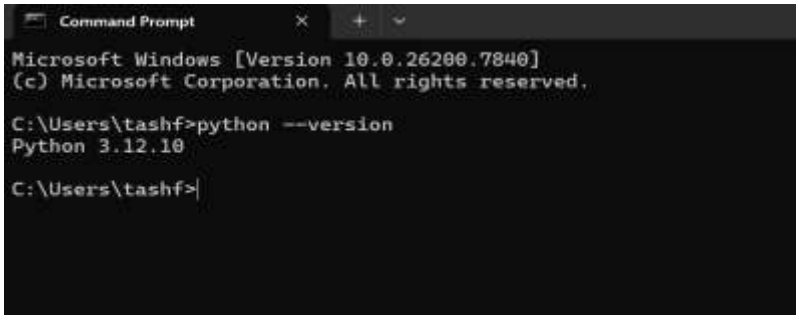


- VS CODE:



- PyCharm

- Python



After installation, I verified everything.

Part 2: Linux Shell Scripting Fundamentals

I used Debian through WSL (Windows Subsystem for Linux) for all of these tasks. Everything was done inside the Debian terminal.

Task 2 – Basic Command Line Operations

Step 1 – Create Working Directory

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ mkdir week1
Tashfeen@DESKTOP-PM7U9BH:~$ cd week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ pwd
/home/Tashfeen/week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

Explanation:

I created a directory named week1, navigated into it, and confirmed my location.

Commands Used:

- mkdir week_one → Creates a new directory.
- cd week_one → Changes directory.
- pwd → Shows current path.

Creating Directory Structure

The mkdir command creates multiple directories and ls lists directory contents.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ mkdir dir1 dir2 dir3
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
dir1  dir2  dir3
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

Creating Files Using touch

The touch command creates empty files and ls -l displays detailed file information.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ cd dir1
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ touch file1.txt file2.txt file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ ls -l
total 0
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file1.txt
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file2.txt
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$
```

Commands Demonstration

1.pwd Command

The pwd command displays the current working directory.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ mkdir week1
Tashfeen@DESKTOP-PM7U9BH:~$ cd week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ pwd
/home/Tashfeen/week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

2.cd Command

The cd command is used to change directories.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ mkdir week1
Tashfeen@DESKTOP-PM7U9BH:~$ cd week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ pwd
/home/Tashfeen/week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

3. ls Command (-l and-a)

The ls command lists files and directories, while -l shows details and -a shows hidden files.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ ls -l
total 0
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file1.txt
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file2.txt
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:27 file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ ls -a
.  ..  file1.txt  file2.txt  file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$
```

4. cat Command

The cat command displays file contents.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ cat dir1/file1.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

5. sudo Command (Safe)

The sudo command executes commands with administrative privileges.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ sudo ls
[sudo] password for Tashfeen:
dir1  dir2  dir3
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

6. man Command

The man command displays the manual page of a command.

```
LS(1)                                User Commands                                LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about  the FILEs (the current directory by default).  Sort entries alphabetically if none of -cftuvSUX nor
       --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~

 Manual page ls(1) line 1 (press h for help or q to quit)
```

7. mkdir Command

The mkdir command creates a new directory.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ mkdir test1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
dir1  dir2  dir3  test1
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

8. rmdir Command

The rmdir command removes an empty directory.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ rmdir test1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
dir1  dir2  dir3
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

9. rm Command

The rm command deletes files.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls dir1
file1.txt  file2.txt  file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ rm dir1/file3.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls dir1
file1.txt  file2.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

10. cp Command

The cp command copies files.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ cp dir1/file1.txt copy.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
copy.txt  dir1  dir2  dir3
```

11. mv Command

The mv command moves or renames files.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ mv copy.txt new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
dir1  dir2  dir3  new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

12. Redirection (>)

The > operator writes output to a file and overwrites existing content.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ cd dir1
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ echo 'Hello World' > file.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ cat file.txt
Hello World
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$
```

13. Append (>>)

The >> operator appends data to an existing file.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ echo "Hello World" > file.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ cat file.txt
Hello World
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$  echo "Hello Universe" >> file.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ cat file.txt
Hello World
Hello Universe
Tashfeen@DESKTOP-PM7U9BH:~/week1/dir1$ |
```

TASK 3

Create sample.txt

The touch command creates a new file.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ touch sample.txt
Tashfeen@DESKTOP-PM7U9BH:~$ nano sample.txt
Tashfeen@DESKTOP-PM7U9BH:~$ ls
1  sample.txt  week  week1
Tashfeen@DESKTOP-PM7U9BH:~$
```

Add 20 Lines

```
GNU nano 8.4                                    sample.txt *
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
Line 12
Line 13
Line 14
Line 15
Line 16
Line 17
Line 18
Line 19
Line 20
```

Viewing Commands

file Command

The file command determines the type of a file.

```
Tashfeen@DESKTOP-PM7U9BH:~$ file sample.txt
sample.txt: ASCII text
Tashfeen@DESKTOP-PM7U9BH:~$
```

cat Command

The cat command shows contents of file.

```
Tashfeen@DESKTOP-PM7U9BH:~$ cat sample.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
Line 12
Line 13
Line 14
Line 15
Line 16
Line 17
Line 18
Line 19
Line 20
```

More command

```
Tashfeen@DESKTOP-PM7U9BH:~$ more sample.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
Line 12
Line 13
Line 14
Line 15
Line 16
Line 17
Line 18
Line 19
Line 20
```

Less Command

```
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
Line 12
Line 13
Line 14
Line 15
Line 16
Line 17
Line 18
Line 19
Line 20

~
~
~
~
~
~
~
~
~
~
~
(END)
```

Head and head-5 Command

```
Tashfeen@DESKTOP-PM7U9BH:~$ head sample.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Tashfeen@DESKTOP-PM7U9BH:~$ head -5 sample.txt
Line 1
Line 2
Line 3
Line 4
Line 5
Tashfeen@DESKTOP-PM7U9BH:~$
```

Alias Command

The alias command creates a shortcut for an existing command.

```
Tashfeen@DESKTOP-PM7U9BH:~$ alias showfile="cat sample.txt"
Tashfeen@DESKTOP-PM7U9BH:~$ showfile
Line 1
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
Line 8
Line 9
Line 10
Line 11
Line 12
Line 13
Line 14
Line 15
Line 16
Line 17
Line 18
Line 19
Line 20

Tashfeen@DESKTOP-PM7U9BH:~$
```

TASK 4

System & Process Commands

1. shutdown Command

The shutdown command is used to safely power off the system.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ cd week1
Tashfeen@DESKTOP-PM7U9BH:~/week1$ shutdown --help
shutdown [OPTIONS...] [TIME] [WALL...]

Shut down the system.

Options:
     --help      Show this help
  -H --halt      Halt the machine
  -P --poweroff  Power-off the machine
  -r --reboot    Reboot the machine
  -h             Equivalent to --poweroff, overridden by --halt
  -k             Don't halt/power-off/reboot, just send warnings
     --no-wall   Don't send wall message before halt/power-off/reboot
  -c             Cancel a pending shutdown
     --show      Show pending shutdown

This is a compatibility interface, please use the more powerful 'systemctl halt',
'systemctl poweroff', 'systemctl reboot' commands instead.

See the shutdown(8) man page for details.
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

2. restart Command

The reboot command restarts the system.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ reboot --help
reboot [OPTIONS...] [ARG]

Reboot the system.

Options:
     --help      Show this help
     --halt      Halt the machine
  -p --poweroff  Switch off the machine
     --reboot    Reboot the machine
  -f --force     Force immediate halt/power-off/reboot
  -w --wtmp-only Don't halt/power-off/reboot, just write wtmp record
  -d --no-wtmp   Don't write wtmp record
     --no-wall   Don't send wall message before halt/power-off/reboot

This is a compatibility interface, please use the more powerful 'systemctl reboot' command instead.

See the halt(8) man page for details.
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

3. touch Command

The touch command creates an empty file.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ touch newfile.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls
dir1  dir2  dir3  newfile.txt  new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

4. nano Command

The nano command opens a file in a text editor.

5. bash Command

The bash command starts a new Bash shell session.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ bash
Tashfeen@DESKTOP-PM7U9BH:~/week1$ exit
exit
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

6. sh Command

The sh command starts a shell interpreter session.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ sh
$
```

7. chmod Command

The chmod command changes file permissions.

Before Permission Change

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls -l new.txt
-rw-r--r-- 1 Tashfeen Tashfeen 0 Mar  1 16:51 new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

Change Permission

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ chmod 777 new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ls -l new.txt
-rwxrwxrwx 1 Tashfeen Tashfeen 0 Mar  1 16:51 new.txt
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

8. ps Command

The ps command displays running processes.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ps
    PID TTY          TIME CMD
    901 pts/2    00:00:00 bash
    923 pts/2    00:00:00 ps
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

9. kill Command

Background Process Start

Last number = Process ID (PID)

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ sleep 1000 &
[1] 924
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

Process Check

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ps
    PID TTY          TIME CMD
    901 pts/2    00:00:00 bash
    924 pts/2    00:00:00 sleep
    925 pts/2    00:00:00 ps
Tashfeen@DESKTOP-PM7U9BH:~/week1$ |
```

Kill Process

The kill command terminates a running process using its process ID.

```
Tashfeen@DESKTOP-PM7U9BH:~/week1$ kill 924
Tashfeen@DESKTOP-PM7U9BH:~/week1$ ps
    PID TTY          TIME CMD
    901 pts/2    00:00:00 bash
    926 pts/2    00:00:00 ps
[1]+  Terminated              sleep 1000
Tashfeen@DESKTOP-PM7U9BH:~/week1$
```

TASK 5

Package Management

1. Installed Packages

Displays installed packages.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/half-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                      Version                  Architecture Description
+++-==========================-========================-============-==================================================>
ii  adduser                   3.152                    all          add and remove users and groups
ii  apt                       3.0.3                    amd64        commandline package manager
ii  apt-utils                 3.0.3                    amd64        package management related utility programs
ii  base-files                13.8+deb13u2             amd64        Debian base system miscellaneous files
ii  base-passwd               3.6.7                    amd64        Debian base system master password and group files
ii  bash                      5.2.37-2+b5              amd64        GNU Bourne Again SHell
ii  bsdutils                  1:2.41-5                 amd64        basic utilities from 4.4BSD-Lite
ii  coreutils                 9.7-3                    amd64        GNU core utilities
ii  cpio                      2.15+dfsg-2              amd64        GNU cpio -- a program to manage archives of files
ii  cron                      3.0pl1-197               amd64        process scheduling daemon
ii  cron-daemon-common        3.0pl1-197               all          process scheduling daemon's configuration files
ii  dash                      0.5.12-12                amd64        POSIX-compliant shell
ii  dbus                      1.16.2-2                 amd64        simple interprocess messaging system (system message bus)
ii  dbus-bin                  1.16.2-2                 amd64        simple interprocess messaging system (command line utiliti>
ii  dbus-daemon               1.16.2-2                 amd64        simple interprocess messaging system (reference message bu>
ii  dbus-session-bus-common   1.16.2-2                 all          simple interprocess messaging system (session bus configur>
ii  dbus-system-bus-common    1.16.2-2                 all          simple interprocess messaging system (system bus configura>
ii  debconf                   1.5.91                   all          Debian configuration management system
ii  debconf-i18n              1.5.91                   all          full internationalization support for debconf
ii  debian-archive-keyring    2025.1                   all          OpenPGP archive certificates of the Debian archive
ii  debianutils               5.23.2                   amd64        Miscellaneous utilities specific to Debian
ii  dhcpcd-base               1:10.1.0-11+deb13u1      amd64        DHCPv4 and DHCPv6 dual-stack client (binaries and exit hoo>
ii  diffutils                 1:3.10-4                 amd64        File comparison utilities
ii  dmidecode                 3.6-2                    amd64        SMBIOS/DMI table decoder
ii  dpkg                      1.22.21                  amd64        Debian package management system
ii  e2fsprogs                 1.47.2-3+b3              amd64        ext2/ext3/ext4 file system utilities
ii  fdisk                     2.41-5                   amd64        collection of partitioning utilities
```

2. Update Packages

Updates package list.

```
Tashfeen@DESKTOP-PM7U9B
Tashfeen@DESKTOP-PM7U9BH:~$ sudo apt upgrade
[sudo] password for Tashfeen:
Upgrading:
  base-files  dhcpcd-base  libc-bin   libc6    libcap2-bin  libext2fs2t64  libssl3t64  logsave                  sqv
  bash        e2fsprogs    libc-l10n  libcap2  libcom-err2  libss2         locales     openssl-provider-legacy

Summary:
  Upgrading: 17, Installing: 0, Removing: 0, Not Upgrading: 0
  Download size: 14.3 MB
  Freed space: 5,120 B

Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libc6 amd64 2.41-12+deb13u1 [2,851 kB]
Get:2 http://security.debian.org/debian-security trixie-security/main amd64 openssl-provider-legacy amd64 3.5.4-1~deb13u2 [308 kB]
Get:3 http://security.debian.org/debian-security trixie-security/main amd64 libssl3t64 amd64 3.5.4-1~deb13u2 [2,442 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 base-files amd64 13.8+deb13u3 [73.3 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 bash amd64 5.2.37-2+b7 [1,501 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 libc-bin amd64 2.41-12+deb13u1 [637 kB]
Get:7 http://deb.debian.org/debian trixie/main amd64 logsave amd64 1.47.2-3+b7 [24.9 kB]
Get:8 http://deb.debian.org/debian trixie/main amd64 libext2fs2t64 amd64 1.47.2-3+b7 [213 kB]
Get:9 http://deb.debian.org/debian trixie/main amd64 e2fsprogs amd64 1.47.2-3+b7 [591 kB]
Get:10 http://deb.debian.org/debian trixie/main amd64 libcap2 amd64 1:2.75-10+b3 [28.8 kB]
Get:11 http://deb.debian.org/debian trixie/main amd64 sqv amd64 1.3.0-3+b2 [661 kB]
Get:12 http://deb.debian.org/debian trixie/main amd64 dhcpcd-base amd64 1:10.1.0-11+deb13u2 [201 kB]
Get:13 http://deb.debian.org/debian trixie/main amd64 libc-l10n all 2.41-12+deb13u1 [738 kB]
Get:14 http://deb.debian.org/debian trixie/main amd64 locales all 2.41-12+deb13u1 [3,925 kB]
Get:15 http://deb.debian.org/debian trixie/main amd64 libcap2-bin amd64 1:2.75-10+b3 [36.4 kB]
Get:16 http://deb.debian.org/debian trixie/main amd64 libcom-err2 amd64 1.47.2-3+b7 [25.0 kB]
Get:17 http://deb.debian.org/debian trixie/main amd64 libss2 amd64 1.47.2-3+b7 [29.7 kB]
Fetched 14.3 MB in 5s (2,666 kB/s)
Preconfiguring packages ...
(Reading database ... 8742 files and directories currently installed.)
Preparing to unpack .../libc6_2.41-12+deb13u1_amd64.deb ...
```

```
Unpacking sqv (1.3.0-3+b2) over (1.3.0-3) ...
Setting up sqv (1.3.0-3+b2) ...
(Reading database ... 8743 files and directories currently installed.)
Preparing to unpack .../0-dhcpcd-base_1%3a10.1.0-11+deb13u2_amd64.deb ...
Unpacking dhcpcd-base (1:10.1.0-11+deb13u2) over (1:10.1.0-11+deb13u1) ...
Preparing to unpack .../1-libc-l10n_2.41-12+deb13u1_all.deb ...
Unpacking libc-l10n (2.41-12+deb13u1) over (2.41-12) ...
Preparing to unpack .../2-locales_2.41-12+deb13u1_all.deb ...
Unpacking locales (2.41-12+deb13u1) over (2.41-12) ...
Preparing to unpack .../3-libcap2-bin_1%3a2.75-10+b3_amd64.deb ...
Unpacking libcap2-bin (1:2.75-10+b3) over (1:2.75-10+b1) ...
Preparing to unpack .../4-libcom-err2_1.47.2-3+b7_amd64.deb ...
Unpacking libcom-err2:amd64 (1.47.2-3+b7) over (1.47.2-3+b3) ...
Preparing to unpack .../5-libss2_1.47.2-3+b7_amd64.deb ...
Unpacking libss2:amd64 (1.47.2-3+b7) over (1.47.2-3+b3) ...
Setting up libc-l10n (2.41-12+deb13u1) ...
Setting up libcom-err2:amd64 (1.47.2-3+b7) ...
Setting up locales (2.41-12+deb13u1) ...
Generating locales (this might take a while)...
  en_US.UTF-8... done
Generation complete.
Setting up libcap2-bin (1:2.75-10+b3) ...
Setting up libss2:amd64 (1.47.2-3+b7) ...
Setting up dhcpcd-base (1:10.1.0-11+deb13u2) ...
Installing new version of config file /etc/dhcpcd.conf ...
Setting up logsave (1.47.2-3+b7) ...
Setting up e2fsprogs (1.47.2-3+b7) ...
Processing triggers for debianutils (5.23.2) ...
Processing triggers for libc-bin (2.41-12+deb13u1) ...
Processing triggers for systemd (257.9-1~deb13u1) ...
Tashfeen@DESKTOP-PM7U9BH:~$
```